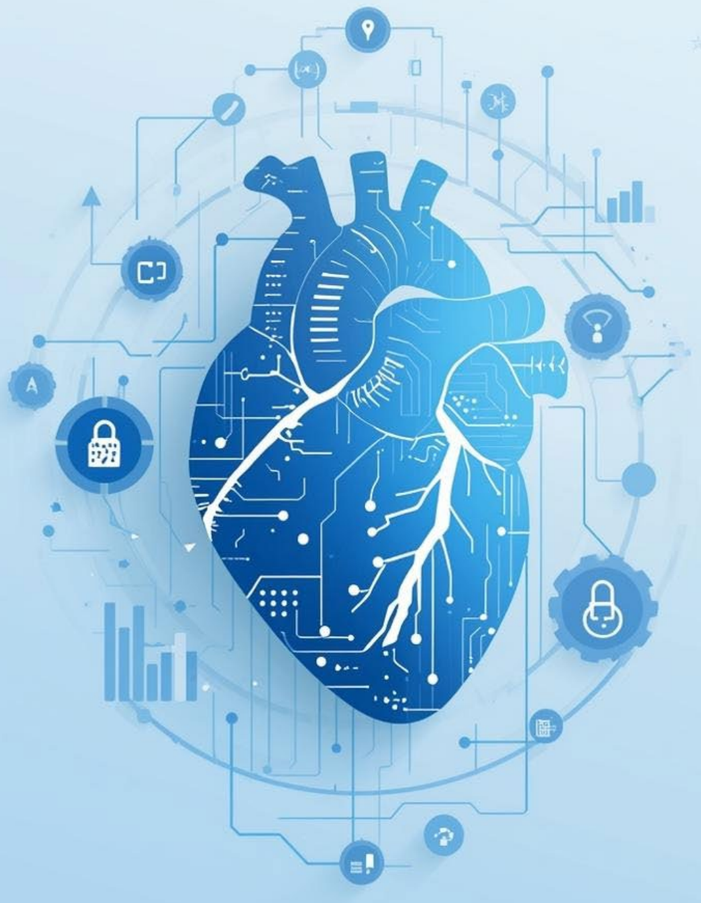
SMARTHEART TECHNOLOGIES

Machine Learning for Heart Disease



Presented by: Allen George, Sinneh, Ibrahim, Makayla Pelt

Importance of Prediction

Clinical and operational relevance

What is being predicted?

SmartHeart predicts five severity levels (0–4) instead of just “disease” vs “no disease”

This helps separate low-risk, moderate, and high-risk patients more clearly

Why severity prediction matters for clinicians

Treatment decisions depend on how severe the disease is, not just whether it exists

Misclassifying even one level can change hospitalization, medication, or monitoring plans

Better severity prediction supports earlier interventions and reduces emergency events

Why machine learning is needed

Traditional methods rely on manual reading of ECG, blood pressure, cholesterol, etc.

Important patterns in the data can be subtle or complex and easy to overlook

Machine learning can find hidden relationships in structured patient data and support faster, more accurate decisions.

Business Impact of SmartHeart

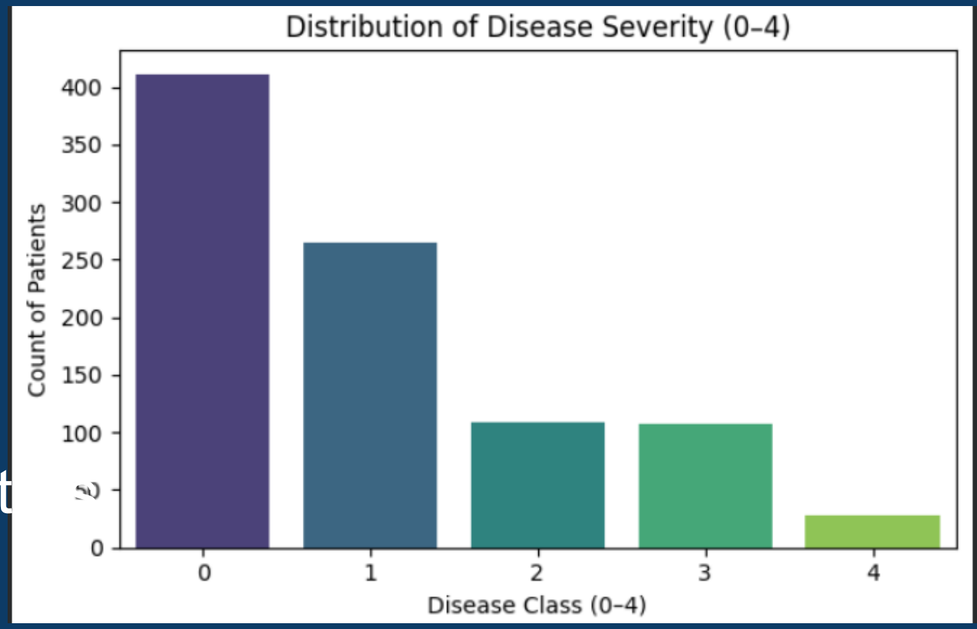
Enhancing healthcare efficiency and outcomes

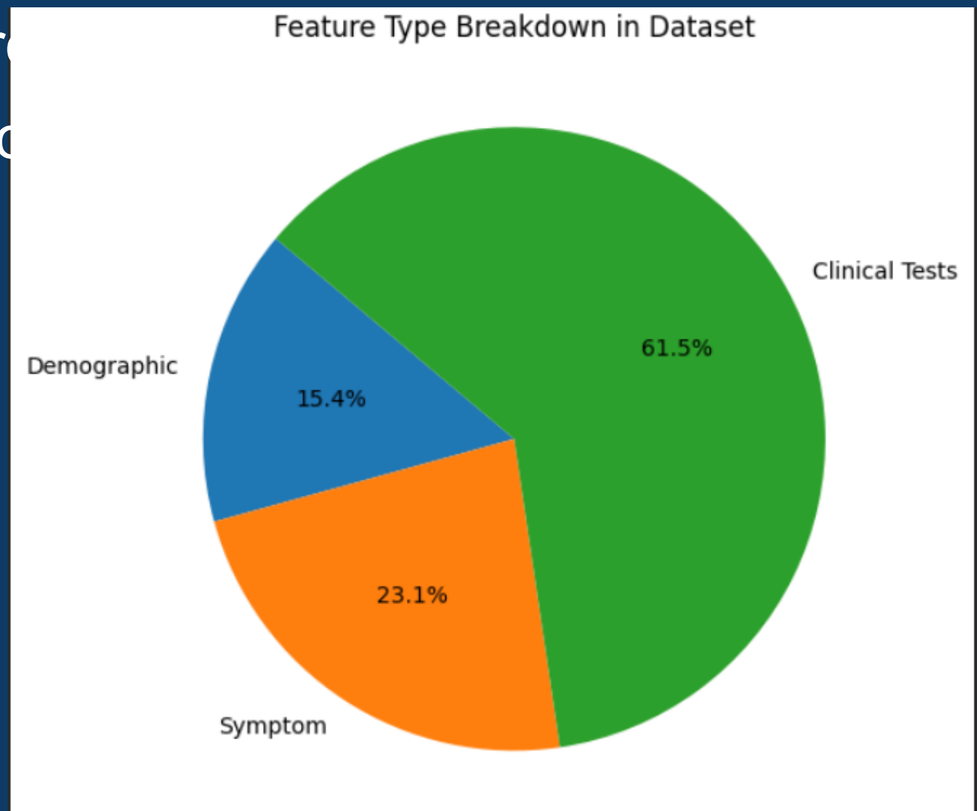
Hospitals and clinicians	Insurance and cost saving	Digital health and inovation
• Reduces unnecessary hospitalizations	• Early detection lowers long-term treatment costs	• Integrates into telehealth and remote monitoring systems
• Improves triage and care planning	• Fewer emergency procedures and repeat testing	• Provides scalable AI tools for preventative care
• Supports clinicians with reliable, real-time decision guidance	• Improves risk assessment for value-based care models	• Strengthens healthcare organizations' data-driven decision making

UCI Heart Disease Dataset

Overview of Key Features

- 920 patient records collected from multiple cardiac cent
  - 16 input features + 5-level target variable (num: 0–4)
- Mix of demographic, symptom, and clinical test featur
- Data is structured for multi-level severity classification, n
  just binary disease prediction





Dataset Key Features

Sex

Sex indicates the gender of the patient, which plays a role in heart disease risk factors.

Age

Age represents the patient's age in years, influencing the likelihood of heart disease development.

Resting BP

Trestbps indicates the patient's resting blood pressure, measured in mm Hg, crucial for cardiovascular assessment.

Exercise Angina

Exercise-induced angina presence indicates whether the patient experiences chest pain during physical activity.

Max Heart Rate

Thalch reflects the maximum heart rate achieved during exercise, vital for assessing cardiac capacity and health.

Cholesterol

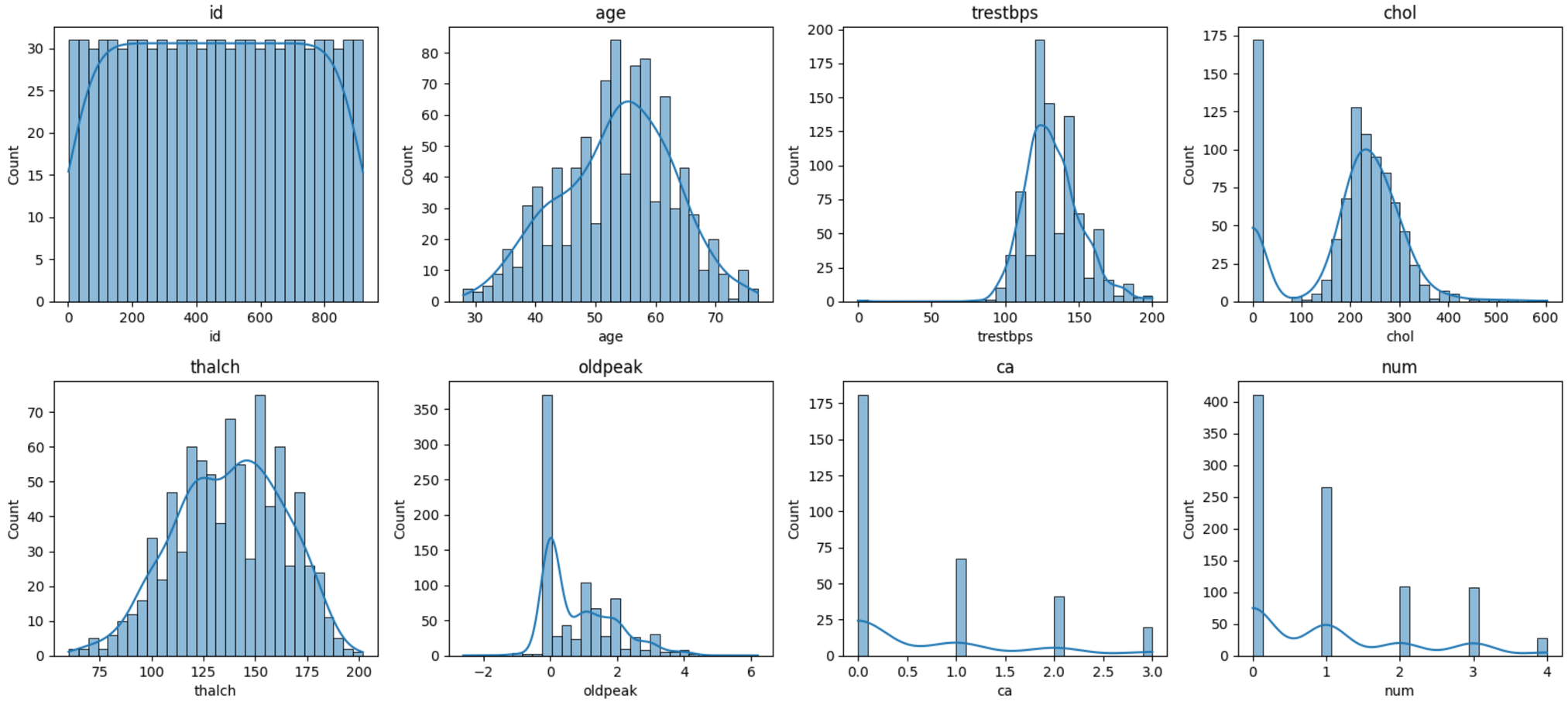
Chol represents the serum cholesterol level, measured in mg/dl, important for evaluating heart disease risk.

Chest Pain

Chest pain type is categorized into four distinct types, reflecting varying symptomatology during episodes.

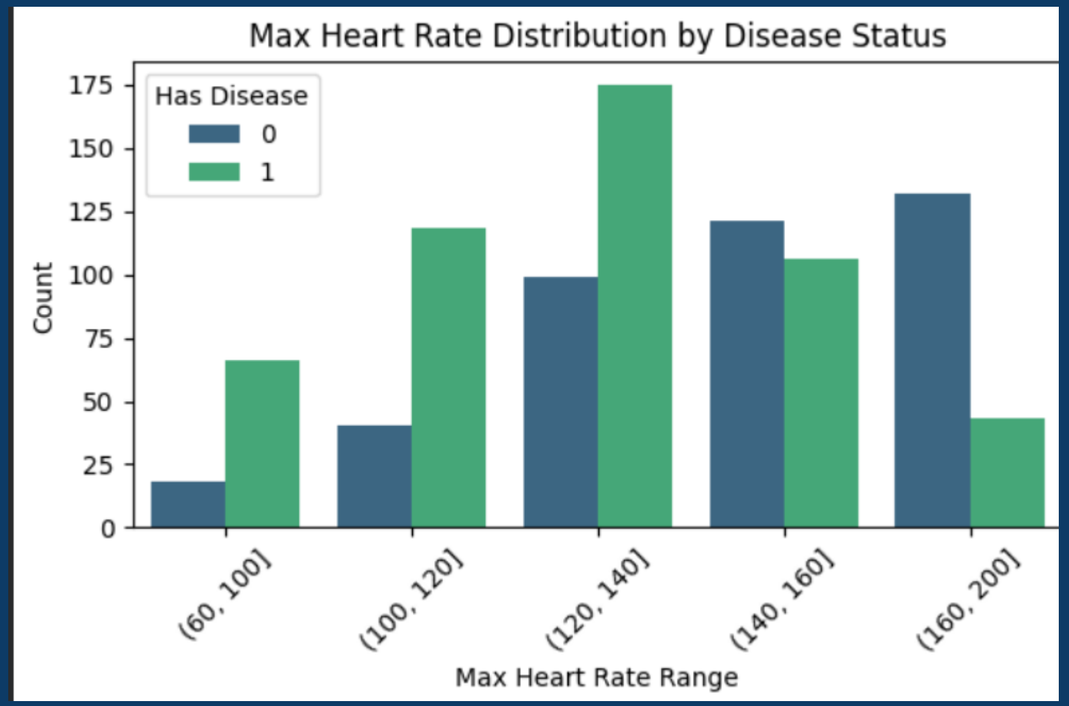
EDA Visualizations Insights

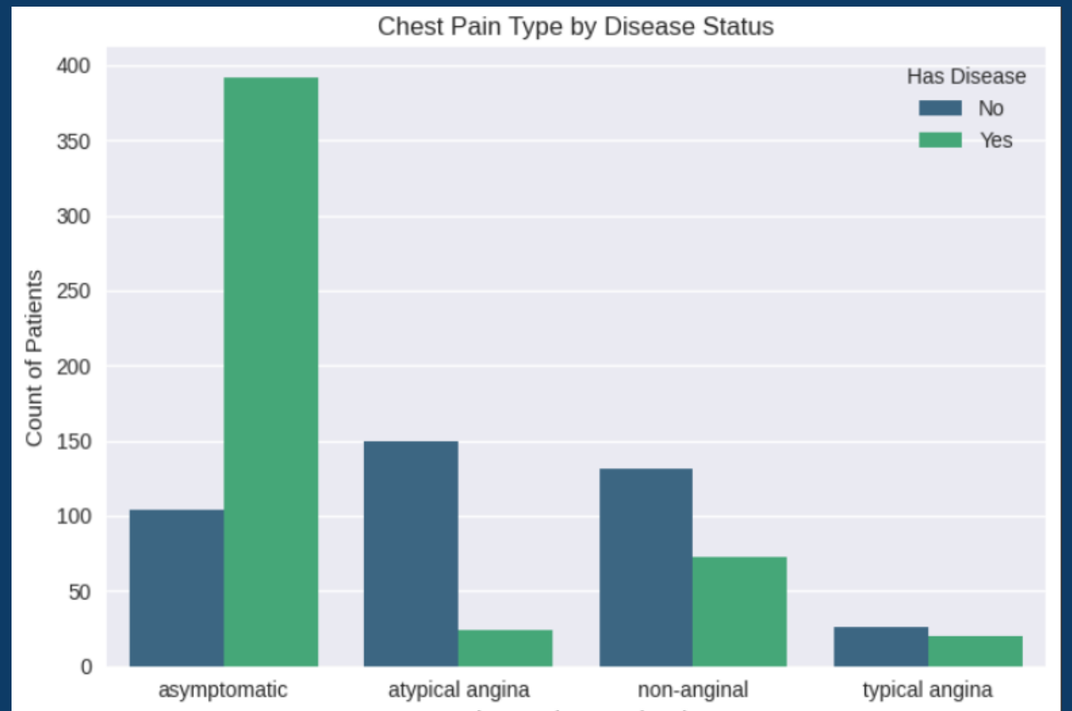
Understanding the dataset features



Exploratory Data Analysis

Key Findings and Insights

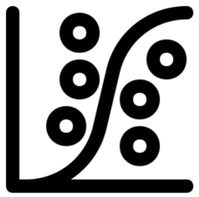




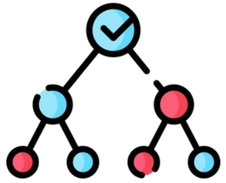
- Most patients between the ages of 45-65, majority male
- Target variable is slightly imbalanced, justifying th use of SMOTE
- Patients with disease tend to have lower max hea rate, higher oldpeak and more major vessels
- Clear seperation in chest pain when looking at patients with and without heart disease

Models Employed in SmartHeart

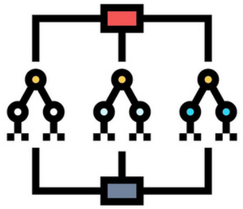
Selected Algorithms and Rationale



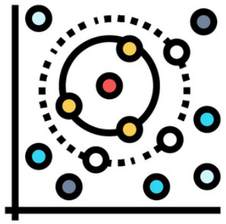
Logistic Regression



Decision Trees



Random Forest



K-Nearest Neighbor

Why These Models?

- Mix of interpretable and high-accuracy algorithms
  - Covers both linear and non-linear relationships
  - Supports comparison across multiple modeling approaches
    - Well-suited for medical risk prediction tasks

Modeling Pipeline Overview

Steps for most Effective Prediction (Pre-Processing)

- Cleaned, imputed and onehot encoded all variables
- Train/ Test Split and Scaling
- Handling Imbalance (Smote and Oversampling)
- Logistic Regression, Decision Tree, KNN, Random Forest
- Tuning, used GridSearchCV to optimize each model

Performance Metrics Overview

Evaluating Model Effectiveness

Key Metrics Used

- Accuracy – proportion of total correct predictions
- Precision – how many predicted positives were truly positive
- Recall (Sensitivity) – how many actual positives were correctly identified
- F1-Score – balances precision and recall
- ROC-AUC – measures overall ability to distinguish between classes

Why These Metrics?

- Ensures balanced evaluation for imbalanced medical data
- Prioritizes Recall to reduce false negatives (critical in healthcare)
- ROC-AUC gives threshold-independent comparison of models

Model Performance Overview

Comparing Algorithm Effectiveness

TUNED MODEL SUMMARY

	model	accuracy	precision	recall	f1	roc_auc
0	Logistic Tuned	0.826087	0.830189	0.862745	0.846154	0.910622
2	KNN Tuned	0.851449	0.863636	0.869281	0.866450	0.907806
3	Random Forest Tuned	0.826087	0.852349	0.830065	0.841060	0.902811
1	Decision Tree Tuned	0.778986	0.823944	0.764706	0.793220	0.819358

- KNN Tuned delivered the highest overall accuracy and F1-score.
- Logistic Regression Tuned achieved the highest ROC-AUC (best class separation).
- Random Forest Tuned provided a strong balance of precision, recall, and stability.
- Decision Tree Tuned performed reasonably but remained the weakest model

Key Insights and Future Work

- The strongest indicators were maximum heart rate, oldpeak, and the number of major vessels while cholesterol and resting blood pressure contributed less.

- KNN delivered the best accuracy and F1-score, and Logistic Regression showed the most apparent class separation, proving that using multiple models strengthens overall insights.

Conclusion

SmartHeart shows how machine learning can improve heart disease assessment by pred severity instead of just providing a yes/ no diagnosis. This gives clinicians more useful information for triage, treatment, and follow-up. While the system can still be refined, our results highlight the value of combining AI with healthcare to support earlier detection and better clinical decisions. With further development and real-world use, SmartHeart has the potential to become a valuable tool in modern cardiovascular care.